THEORY OF MACHINES - I

II B.TECH II Semester: MECHANICAL ENGINEERING

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME19	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OVER VIEW:

This course will deal with kinematic analysis of mechanisms and machines. It covers basic concepts and principles of various mechanisms, analysis of velocity and acceleration of different mechanisms, study of steering gear mechanisms, gears, gear trains, cams and followers.

COURSE OUTCOMES:

The course should enable the students to:

1. Outline basic concepts and principles of mechanisms for doing useful work.
2. Solve velocity and acceleration in different mechanisms.
3. Analyze straight line and steering gear mechanisms.
4. Select the gears and gear trains for transmission of motion.
5. Design cam profile for different follower motions.

UNIT-I INTRODUCTION

Introduction: Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematic pairs – Types of constrained motion – kinematic chain – Mechanism-Machine-Structure – Kleins criteria,

Degrees of freedom, Grubler's Criterion, Grashof's law – inversion of mechanism – inversions of four bar chain, single and double slider crank chains. Spatial mechanisms, introduction to the kinematics of spatial mechanisms. Limit positions, Mechanical Advantage, Transmission Angle.

UNIT-II VELOCITY & ACCELERATION ANALYSIS

Velocity in Mechanisms: Relative Velocity Method- Relative velocities of two bodies moving in straight line, velocity of a point on a link by relative velocity method, Velocities in four bar and single slider crank mechanisms – Problems.
Instantaneous Centre Method: Introduction, Velocity of a point on a link, Types of instantaneous centres, Kennedy's theorem, Method of locating instantaneous centres, determination of angular velocities of points and links – Problems.

Acceleration in Mechanisms: Acceleration of a point on a link Acceleration in four bar and single slider crank mechanisms– Problems. Introduction to synthesis of linkages (Dimensional synthesis for motion, function and path generation).

UNIT-III STRAIGHT LINE MOTION & STEERING GEAR MECHANISMS

Straight line motion mechanisms: Types, Peaucellier, Hart's, Scott Russel, Tchebichef's, Watt's, Grasshopper and Robert Mechanisms - Pantograph.
Steering gear mechanisms: Introduction, Condition for correct steering, Davis Steering gear mechanism, Ackerman's steering gear mechanism. Hooke's joint: Single and double Hooke's joint – velocity ratio – application – Problems.

UNIT-IV	GEARS & GEAR TRAINS
Friction wheels and toothed gears – types – spur gear terminology, law of gearing, condition for constant velocity ratio fortransmission of motion, phenomena of interference, condition for minimum number of teeth to avoid interference, expressions for arc of contact, path of contact and contact ratio of pinion and gear. **Gear trains:** Introduction – Types – Simple, compound and reverted gear trains, Epicyclic gear train – Methods of finding train value or velocity ratio – Problems. Introduction to differential gear for an automobile.	
UNIT-V	**CAMS**
Cams: Introduction, classification of followers and cams; Displacement, velocity and acceleration analysis. Construction of motion of the follower for uniform velocity, simple harmonic motion, uniform acceleration and retardation, cycloidal motion. Introduction to pressure angle and under cutting, sizing of cams.	
Text Books:	
1. Theory of Machines / S.S. Rattan /Tata McGraw-Hill Education. 2. Theory of Machines/Thomas Bevan/Pearson/ 3rdEdition	
Reference Books:	
1 Kinematics and Dynamics of machinery/Robert L. Norton/Tata Mc Graw Hill 2 Mechanisms of machines / Cleghorn W.L. / Oxford University Press 3 TheoryofMechanismandMachines/GhoshandMallik/AffiliatedEast-WestPvt.Ltd. 4 Theory of Machines by R.S Khurmi& J.K Gupta, S.Chand Publications.	